

Dhyani Soni

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY

B.S. in Computer Science & AI Minor, Albert Dorman Honors College

GPA: 3.62 (Dean's List)

- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Data Science, Database Systems, Computer Networks, Operating Systems, Cybersecurity, Linear Algebra, AI-Assisted Software Engineering, Data Visualization, OOP
- **Campus Involvement:** Lyra STEM Curriculum Manager, Kids Who Code Treasurer, SHPE, Rewriting the Code (RTC)
- **Certifications:** Machine Learning Foundations - Cornell University

Expected May 2027

Newark, NJ

SKILLS

- **Languages:** Python, SQL, Java, PHP, JavaScript, C/C++, TypeScript
- **Frameworks & Libraries:** React, FastAPI, Flask, Streamlit, TensorFlow, PyTorch, Scikit-learn, YOLO, FAISS, NumPy, Pandas, Matplotlib
- **AI/ML:** LLMs, Multi-Agent Systems, MCP, LangChain, OpenCV, Sentiment Analysis, Pose Estimation, Object Detection (YOLOv5), Vision-Language Embeddings (CLIP)
- **Cloud & DevOps:** AWS, Docker, Terraform, CI/CD, VMware
- **Tools:** Git, Jupyter Notebook, Linux, Tableau, Figma, Postman, MySQL, Anaconda, GitHub Copilot, Selenium

PROFESSIONAL EXPERIENCE

Artificial Intelligence and Machine Learning Fellow

Break Through Tech - Cornell Tech @ American Express

Mar 2026 - Mar 2027

New York City, New York

- Assessing generative **AI product-market fit** by evaluating LLMs, RAG pipelines, and autonomous AI agents to define which applied AI features can realistically **ship to solve diverse business use cases** along with building across the **full ML cycle**
- Directing an AI Studio team to convert synthetic loan data into a **fair-lending decision engine**, benchmarking logistic regression against XGBoost/LightGBM on ROC-AUC and calibration
- Optimizing a custom **profit-versus-inclusion-gap metric** for executive review, **balancing portfolio return with credit access** for thin-file borrowers

Agentic AI Engineering Intern

BFSI-Unit @ Tata Consultancy Services

May 2026 - Aug 2026

Edison, NJ

- Evaluated actions of reviewers doing manual review (2 hours/report/reviewer), identifying bottlenecks creating repetitive rule-checking, and **produced** an automated system running 18 checks across 5 categories, **cutting turnaround to under 15 seconds per report**
- Refocused **Bedrock Agents** with RAG on past reviews to auto-fixes, validation and a **natural language interface** for querying review history and automatically generating an excel report on the results
- Scoping requirements for a serverless **multi-agent expansion** (Lambda, API Gateway, S3, DynamoDB, Terraform) to take the pipeline from single-report automation to broader, more **scalable deployment**

Artificial Intelligence and Machine Learning Research Intern

CNALab @ New Jersey Institute of Technology

Sept 2025 - May 2026

Newark, NJ

- Won **1st Place** at NJIT's [Honors Interdisciplinary Research Forum](#) (40+ competitors); also selected as a **Finalist** for the Undergraduate Research and Innovation Program for the **prototype** and dataset presented
- Developed an AI-powered rehabilitation tool for **patient mobility tracking** in aquatic therapy by implementing pose estimation with **OpenCV** and **CUDA-accelerated GPU processing** including a custom preprocessing pipeline after **consulting clinicians**
- **Improved keypoint detection by 35%** and reduced labeling errors in the training dataset, increasing joint torque and acceleration analysis accuracy underwater

PROJECTS

[Atlas](#) | Interactive 3D Travel Planning Agent & Visualization Platform | [Docker](#), [Python](#), [Three.js](#), [JavaScript](#)

- Built an interactive **3D globe interface** in Three.js with clickable destination pins, turning a flat travel wishlist into an explorable **spatial visualization**
- Modeled each trip as a **digital twin**, with dedicated planning and negotiating **agents recommending** optimal travel windows, lowest-fare tickets, and top-rated attractions all powered by **LangChain**
- Extended this to group trips, **parsing shared chat input** on budget, food, and activity preferences into a consolidated, printable itinerary

[SenseLense](#) | Best AI Analytics Track at HackHers Hackathon (ADP Challenge) | [Python](#), [SQLite](#), [ElevenLabs](#), [Presage](#), [React](#)

- Architected a real-time multimodal (CV + NLP) sentiment analysis pipeline for sales pitch evaluation at **ADP**
- Integrated Presage (emotion detection), ElevenLabs (speech-to-text), and a **timestamped database** to structure client interaction data
- Leveraged Gemini for **contextual mood analysis** to flag **real-time sentiment shifts** during live pitches, **notifying** reps of negative engagement and suggesting **optimized high-impact pitch strategies**

AgentWeave | ML Recommendation & Retrieval System | [Python](#), [FastAPI](#), [YOLOv5](#), [FAISS](#), [CLIP](#), [React](#), [Firebase](#), [Selenium](#)

- Engineered a computer vision recommendation pipeline using YOLOv5 for user detection and CLIP embeddings to **generate outfit matches** by **visual similarity**, body type, weather, and event context
- Deployed FAISS vector search for **low-latency retrieval** over the embedding space, and built a swipe-based React interface to **stream behavioral feedback** to Firebase
- Refined ranking accuracy by clustering user interactions, personalizing recommendations from **live engagement data** rather than static rules